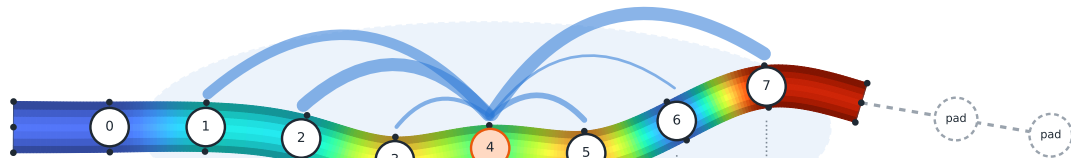


attention window: token 4 sees keys 1 to 7



score $q \cdot k_j / \sqrt{4}$

1.43 1.67 0.14 -2.32 0.65 -0.67 1.69

weight a_j (softmax)

0.22 **0.29** **0.06** **0.01** **0.10** **0.03** **0.29**

value v_j

-0.77	0.27	-0.67	1.00	0.86	0.86	-0.04
-1.52	0.86	-2.00	0.78	1.03	1.90	-0.33
-1.37	-0.45	-0.87	1.62	0.25	1.14	-0.92
-2.84	-1.22	-2.11	2.47	-0.63	2.31	-2.54

$\sum_j a_j v_j$



-0.04
-0.16
-0.69
-1.85

o, the new state of token 4

dashed nodes are padding. They are never attended to, and a padded query keeps only itself so its softmax is defined. Band width is the weight a_j ; they sum to 1.00. One head is drawn, the model has 4.